

Worksheet 8A: Conditional Probability and Bayes' Theorem

Series 8: Probability Deepened
AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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What This Worksheet Is About

In ws06a you calculated the probability of a single event. But most interesting questions involve knowing something already: “What is the probability this email is spam, given it contains the word ‘winner’?” Bayes’ Theorem — arguably the most important formula in machine learning — lets us update beliefs in light of new evidence. This worksheet builds from the definition of conditional probability to the full Naive Bayes classifier.

Part A: Conditional Probability $P(A \mid B)$ (10 problems)

$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
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 $\iff P(A \cap B) = P(A \mid B) \cdot P(B) \quad (\text{multiplication rule})$

1. A bag: 4 red, 6 blue. Draw two without replacement. (a) $P(\text{first red})$? (b) $P(\text{second red} \mid \text{first red})$? (c) $P(\text{both red})$?
2. A class of 30: 18 study maths, 12 French, 6 both. (a) $P(\text{maths})$? (b) $P(\text{French} \mid \text{maths})$? (c) $P(\text{maths} \mid \text{French})$?
3. Roll two dice. $A = \text{“sum is 8”}$, $B = \text{“first die shows 4”}$. (a) $P(A)$? (b) $P(A \mid B)$? (c) Is $P(A \mid B) > P(A)$? Why?
4. A tech company has 260 employees:

	Engineer	Designer	Manager	Total
Female	60	40	20	120
Male	90	20	30	140
Total	150	60	50	260

- Find: (a) $P(\text{Engineer})$ (b) $P(\text{Female} \mid \text{Engineer})$ (c) $P(\text{Engineer} \mid \text{Female})$ (d) $P(\text{Manager} \mid \text{Male})$ (e) Are “Female” and “Engineer” independent?
5. Standard deck (52 cards). $A = \text{“King”}$, $B = \text{“face card”}$ (12 face cards). Find $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \mid B)$, $P(B \mid A)$.
 6. Factory: 5% of components defective. Test correctly identifies defective with probability 0.9; incorrectly flags good with probability 0.02. Let D =defective, $+$ =positive test. Write $P(D)$, $P(D^c)$, $P(+ \mid D)$, $P(+ \mid D^c)$. Then compute $P(D \cap +)$ and $P(D^c \cap +)$.

7. Complement rule: $P(A^c | B) = 1 - P(A | B)$. In problem 4, given the employee is Female, find $P(\text{not Engineer} | \text{Female})$ two ways: from the table, and via the complement rule.
8. Chain rule: $P(A \cap B \cap C) = P(A) \cdot P(B | A) \cdot P(C | A \cap B)$. Find $P(\text{three Aces})$ when drawing 3 cards without replacement.
9. $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$. (a) $P(A | B)$ and $P(B | A)$? (b) Are A and B independent?
10. $P(A | B) = 0.6$, $P(B) = 0.3$. Find $P(A \cap B)$.

Part B: Independence (8 problems)

A and B are **independent** if $P(A | B) = P(A)$, equivalently $P(A \cap B) = P(A) \cdot P(B)$.

11. Fair coin flipped twice. $A = \text{"first Heads"}$, $B = \text{"second Heads"}$. (a) Find $P(A)$, $P(B)$, $P(A \cap B)$. (b) Verify independence.
12. From problem 4: are "Female" and "Designer" independent? Compare $P(F \cap D)$ and $P(F) \cdot P(D)$.
13. Sensor fails with probability 0.01 per day, independently. (a) $P(\text{works 5 days})$? (b) $P(\text{fails at least once in 5 days})$? (c) $P(\text{fails on exactly day 3})$?
14. Three independent events each with $P = 0.5$. $P(\text{all three occur})$?
15. Let $A = \text{"first flip H"}$, $B = \text{"second flip H"}$, $C = \text{"both flips the same"}$. Show A and B are independent, but $P(A \cap B \cap C) \neq P(A)P(B)P(C)$.
16. **Naïve Bayes.** $P(\text{spam}) = 0.3$; $P(\text{winner} | \text{spam}) = 0.8$; $P(\text{free} | \text{spam}) = 0.7$; $P(\text{click} | \text{spam}) = 0.6$. Compute the unnormalised spam score.
17. $P(\text{winner} | \neg \text{spam}) = 0.05$; $P(\text{free} | \neg \text{spam}) = 0.2$; $P(\text{click} | \neg \text{spam}) = 0.3$. Compute the unnormalised not-spam score.
18. Normalise the scores from problems 16–17 to find $P(\text{spam} | \text{winner, free, click})$.

Part C: Law of Total Probability (8 problems)

If $B_1, \dots, B_n$ partition the sample space:

$$P(A) = \sum_{i=1}^n P(A | B_i) \cdot P(B_i)$$

19. Factory: M1 makes 60% (3% defect rate), M2 makes 40% (5% defect rate). (a) Write the probabilities. (b) Find $P(D)$.
20. Email types: 40% work ($P(\text{urgent}) = 0.3$), 35% personal (0.1), 25% spam (0.6). Find $P(\text{"urgent"})$.
21. Disease prevalence 2%; $P(+ | D) = 0.95$; $P(- | D^c) = 0.90$. Compute $P(+)$.
22. Compute $P(-)$ both ways: $1 - P(+)$ and directly.

23. Event cancelled if Clear (0.05), Cloudy (0.30), Rainy (0.90), with probabilities 0.5, 0.3, 0.2. Find $P(\text{cancelled})$.
24. Explain how $P(\mathbf{x}) = \sum_k P(\mathbf{x} \mid Z = k) \cdot P(Z = k)$ (Gaussian Mixture Model) is the law of total probability.
25. Three roads: A (50%, 10% traffic), B (30%, 20%), C (20%, 5%). (a) $P(\text{traffic})$? (b) Set up the calculation for: given traffic, which road most likely?
26. Verify the law of total probability for $P(\text{even}) = \frac{1}{2}$ on a die using partition $\{1, 2\}, \{3, 4\}, \{5, 6\}$ each with probability $\frac{1}{3}$.

Part D: Bayes' Theorem (10 problems)

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}$$

Prior $P(A)$, **likelihood** $P(B \mid A)$, **evidence** $P(B)$, **posterior** $P(A \mid B)$.

27. Factory (problem 19): a component is defective. Find $P(M_1 \mid D)$.
28. Medical test (problem 21): a patient tests positive. Find $P(D \mid +)$. (a) Interpret the answer for a low-prevalence population. (b) Repeat for prevalence 20%.
29. Email (problem 20): an email contains “urgent”. Find $P(\text{spam} \mid \text{urgent})$ and $P(\text{work} \mid \text{urgent})$.
30. Self-driving sensor: $P(+ \mid O) = 0.9$, $P(+ \mid O^c) = 0.01$, $P(O) = 0.05$. Find $P(O \mid +)$.
31. Spam filter: prior $P(\text{spam}) = 0.25$; after word “prize”, posterior = 0.85. After then seeing “winner” ($P(\text{winner} \mid \text{spam}) = 0.9$, $P(\text{winner} \mid \neg \text{spam}) = 0.1$), update again. Describe this sequential process.
32. Archaeologist: prior 60% Roman, 40% Medieval. Radiocarbon gives probability 0.7 for Roman and 0.2 for Medieval. Update.
33. **Monty Hall**. Three doors: one car, two goats. You pick door 1; host opens door 3 (a goat). (a) Prior $P(C_1)$? (b) Using Bayes, compute $P(C_1 \mid H_3)$ and $P(C_2 \mid H_3)$. (c) Should you switch?
34. When is the unnormalised posterior $P(A \mid B) \propto P(B \mid A) \cdot P(A)$ useful?
35. **Naïve Bayes (full)**. $P(\text{class} \mid w_1, \dots, w_n) \propto P(\text{class}) \cdot \prod_i P(w_i \mid \text{class})$. Identify each term. Why is the classifier “naïve”?
36. **Log probabilities**. Using values from problems 16–17, compute log-scores for spam and not-spam (use $\ln$). Same classification?

Part E: Putting It Together (4 problems)

37. A friend says: “ $P(\text{rain} \mid \text{dark clouds}) = 0.98$ — I should cancel.” What would you want to verify?

38. Describe a realistic example of the **base rate fallacy** in medicine, security screening, or social media moderation.
39. (a) Show that the Markov chain update $\boldsymbol{\pi}^{(t+1)} = \boldsymbol{\pi}^{(t)}P$ is the law of total probability applied componentwise. (b) How does the stationary distribution $\boldsymbol{\pi}P = \boldsymbol{\pi}$ relate to long-run probabilities?
40. Write three plain-English sentences summarising Bayes' theorem, then one sentence on why it matters in AI.